

If you could go back in time...

Which year would you go to?

What would you change?

Raise your hand to share your answer!



The Time Machine Interface

```
Stack timeStack;
```

```
void initTimeStack(Stack *timeStack);
```

```
void pushYear(Stack *timeStack, int year);
```

```
int popYear(Stack *timeStack);
```

```
int peekYear(Stack *timeStack);
```

```
bool isTimelineEmpty(Stack *timeStack);
```

```
bool isTimelineFull(Stack *timeStack);
```

The Abstract Time Machine Interface

Stack timeStack;

void initTimeStack(Stack *timeStack);

void pushYear(Stack *timeStack, int year);

int popYear(Stack *timeStack);

int peekYear(Stack *timeStack);

int isTimelineEmpty(Stack *timeStack);

int isTimelineFull(Stack *timeStack);

```
int main() {  
    Stack timeStack;  
    initTimeStack(&timeStack);  
    pushYear(&timeStack, 2026);  
    pushYear(&timeStack, 2015);  
    pushYear(&timeStack, 2006);  
    pushYear(&timeStack, 1996);  
  
    printf("\n--- Returning to present ---\n");  
  
    popYear(&timeStack); // 1996  
    popYear(&timeStack); // back to 2006  
    popYear(&timeStack); // back to 2015  
    popYear(&timeStack); // back to 2026  
}
```

*“We are all connected—by stories, by choices,
and by the people who came before us.”*



Draw your family tree.

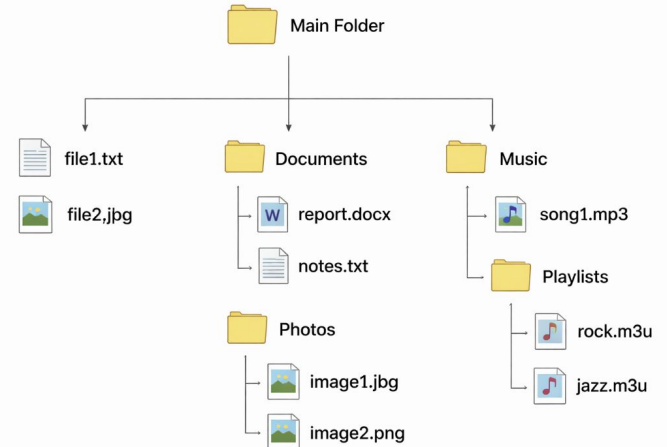
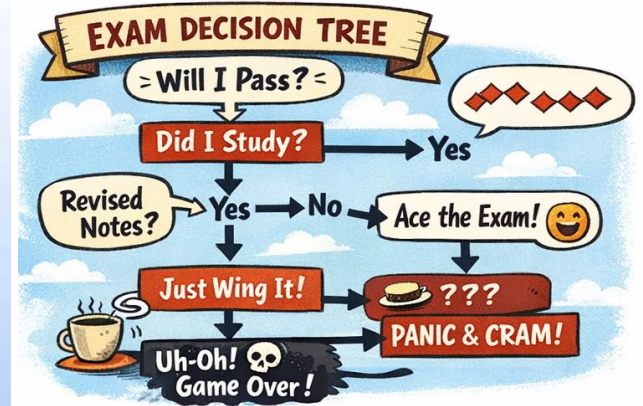
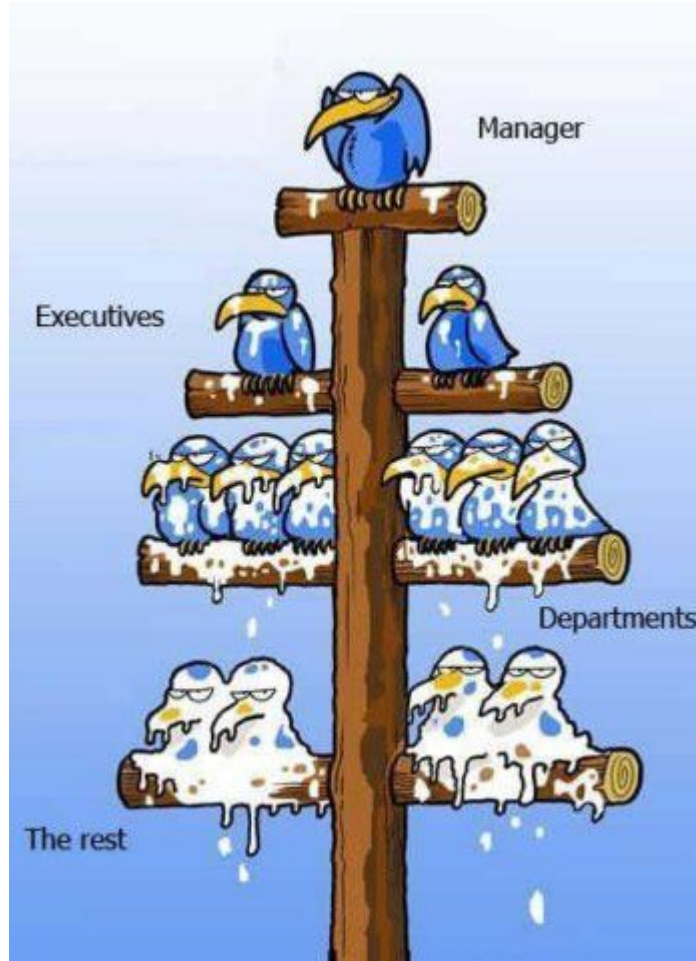
(2–3 minutes activity)

Fun fact: Every person alive today can trace their → **mother** → **mother...**
back to the same woman—who lived in Africa approx. 150,000–200,000
years ago!

Agenda

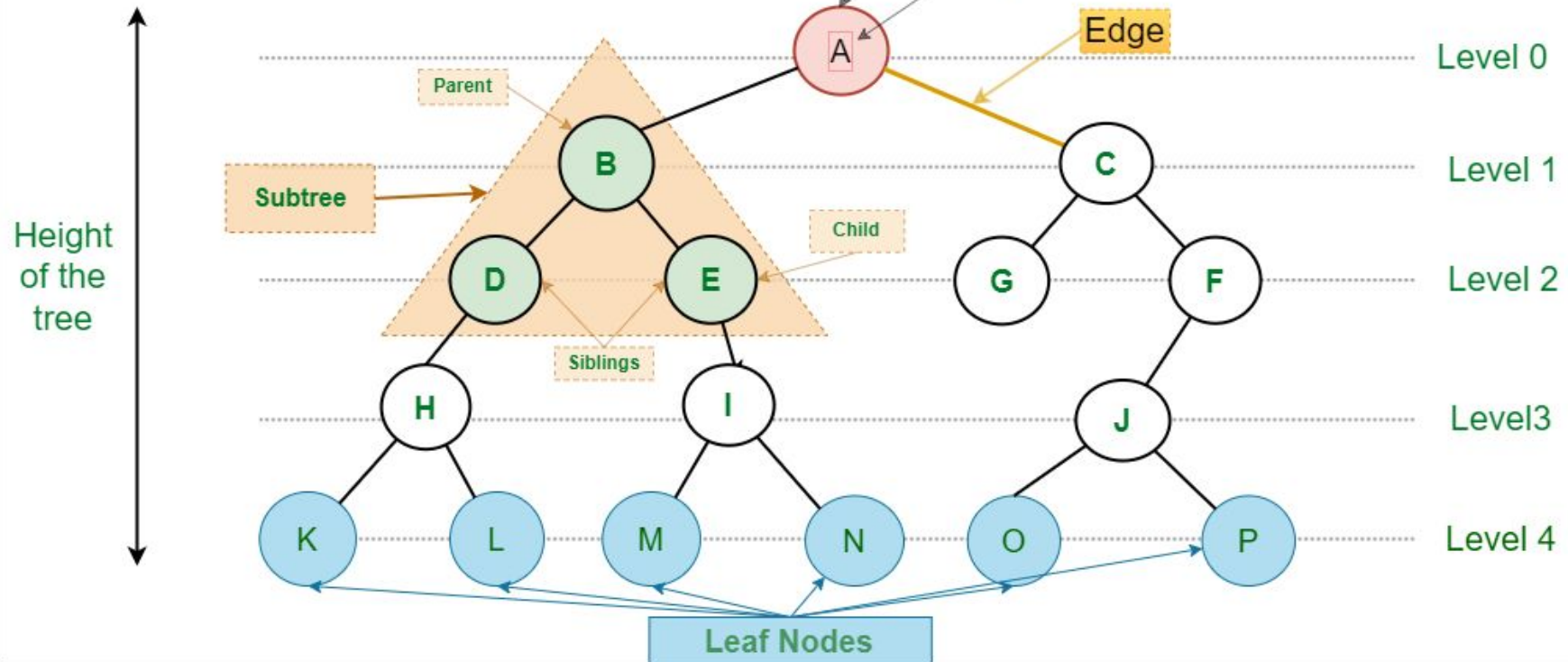
1. Trees
2. Terminologies
3. Binary Trees
4. Implementation

pstree
file tree
website
navigation

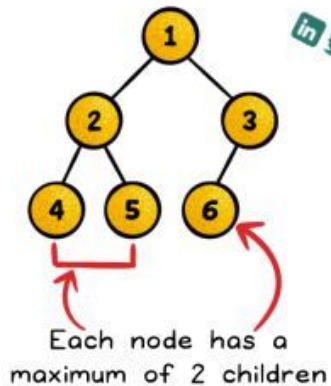


Tree Data Structure

Degree, internal node, external node

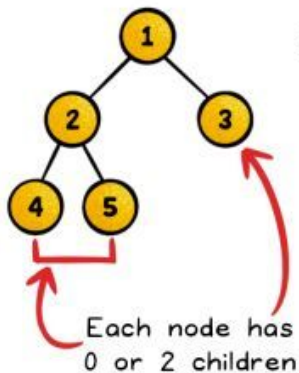


Binary tree

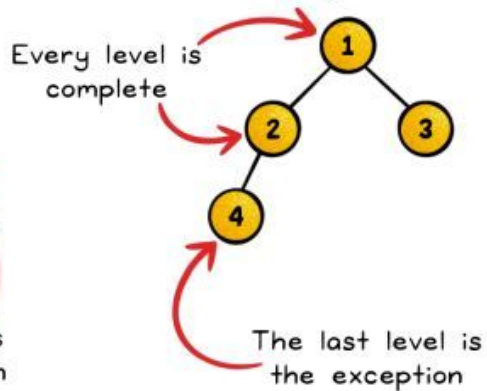


in @NikkiSiapno

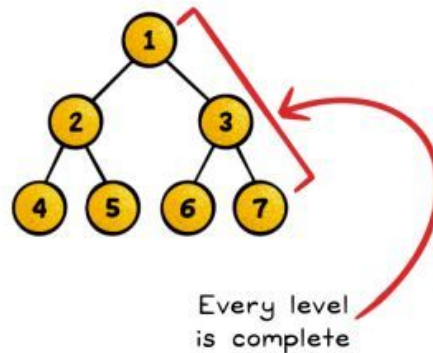
Full binary tree



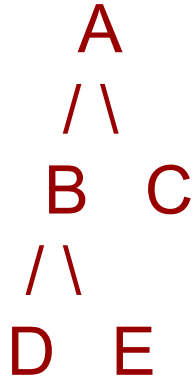
Complete binary tree



Perfect binary tree



Consider the following tree:



What is the degree of node B and the height of the tree?

Given a binary tree with 15 nodes, what is the minimum and maximum possible height of the tree?

A full binary tree has 7 internal nodes. How many total nodes does it have?

Fill in the blank:

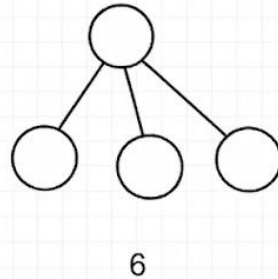
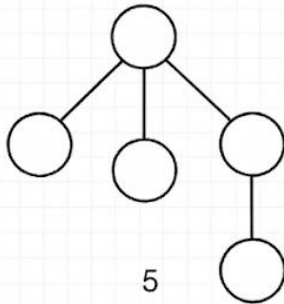
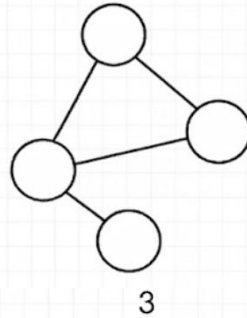
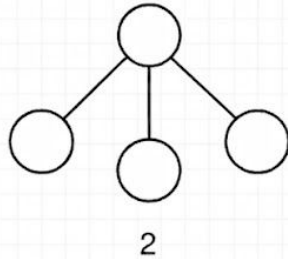
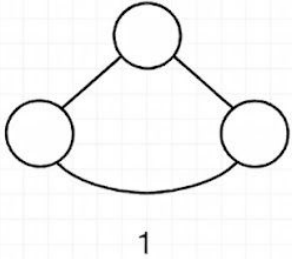
In a binary tree, the maximum number of nodes at level i (root at level 0) is:

max_nodes = _____

AGENDA

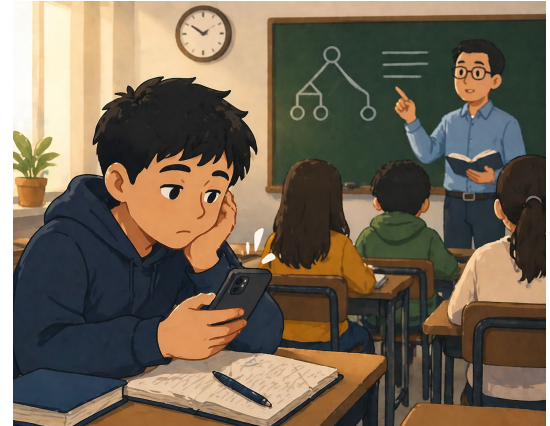
1. Warm up question
2. Representing a binary tree in Computer Memory
 - a. Linked List Representation
 - b. Array Representation
- 3.

Which of the following are trees?

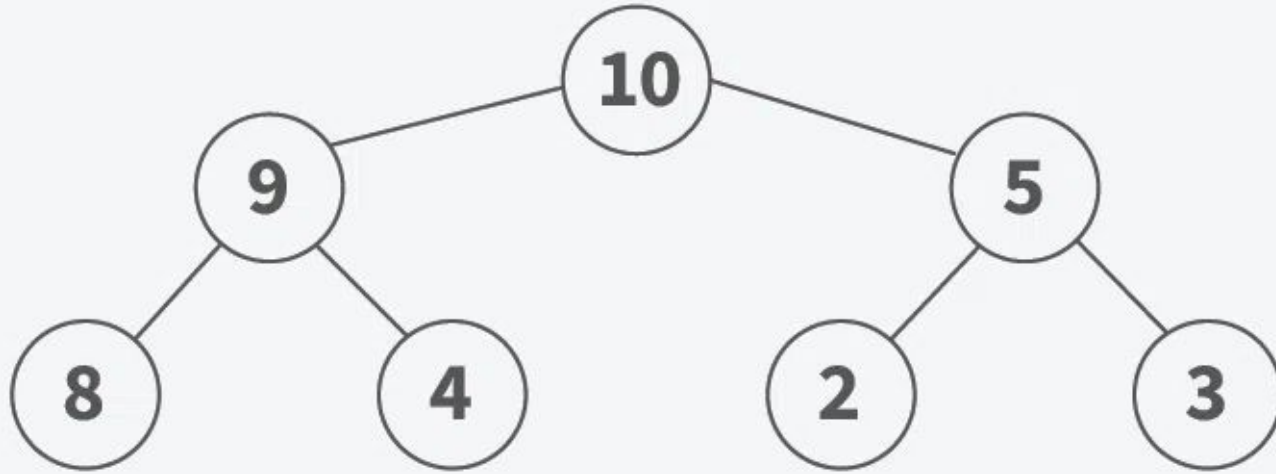


This is a 10-second question... unless you weren't listening.

The tree is simple. The attention wasn't



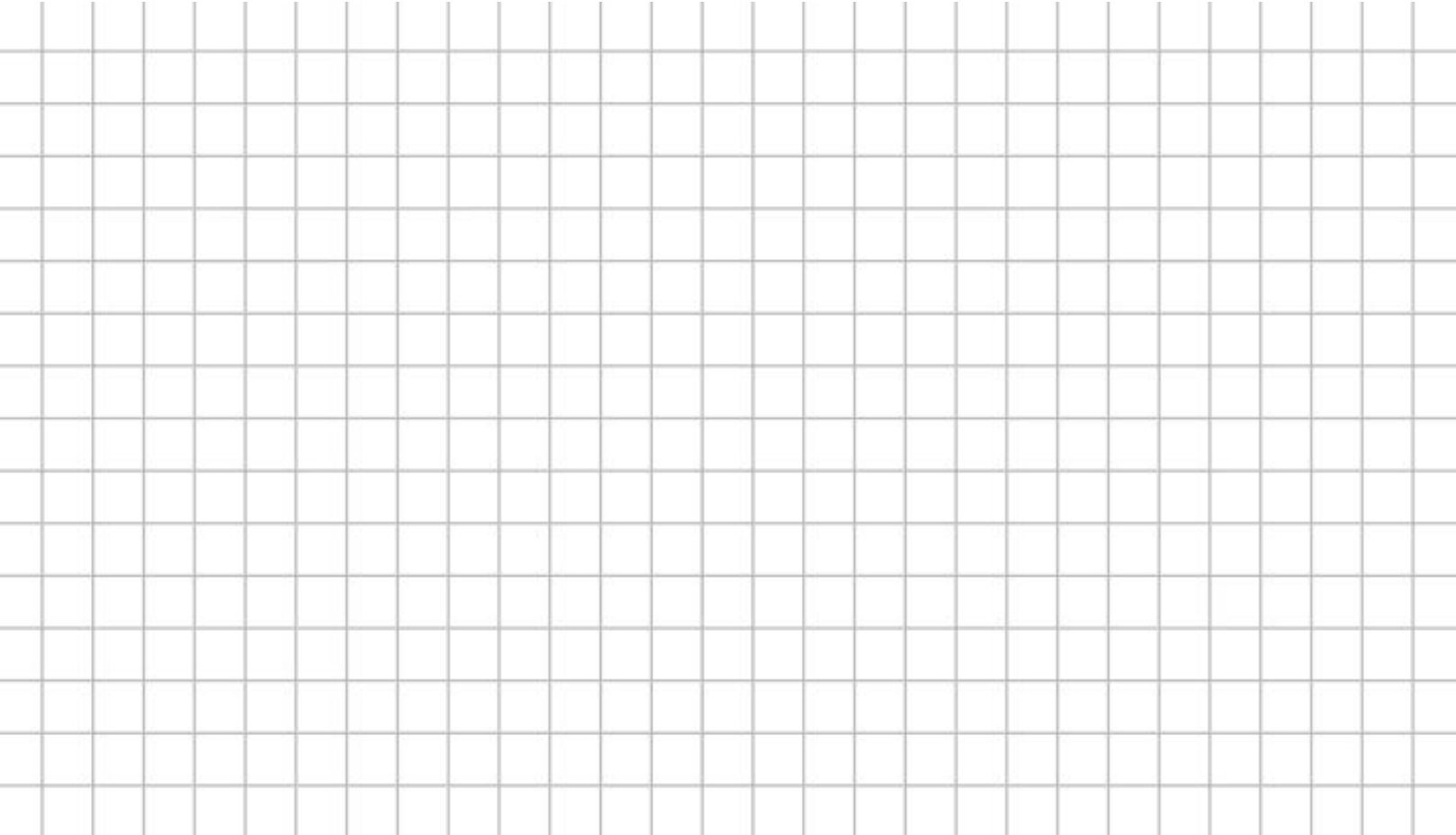
Computers don't understand drawings



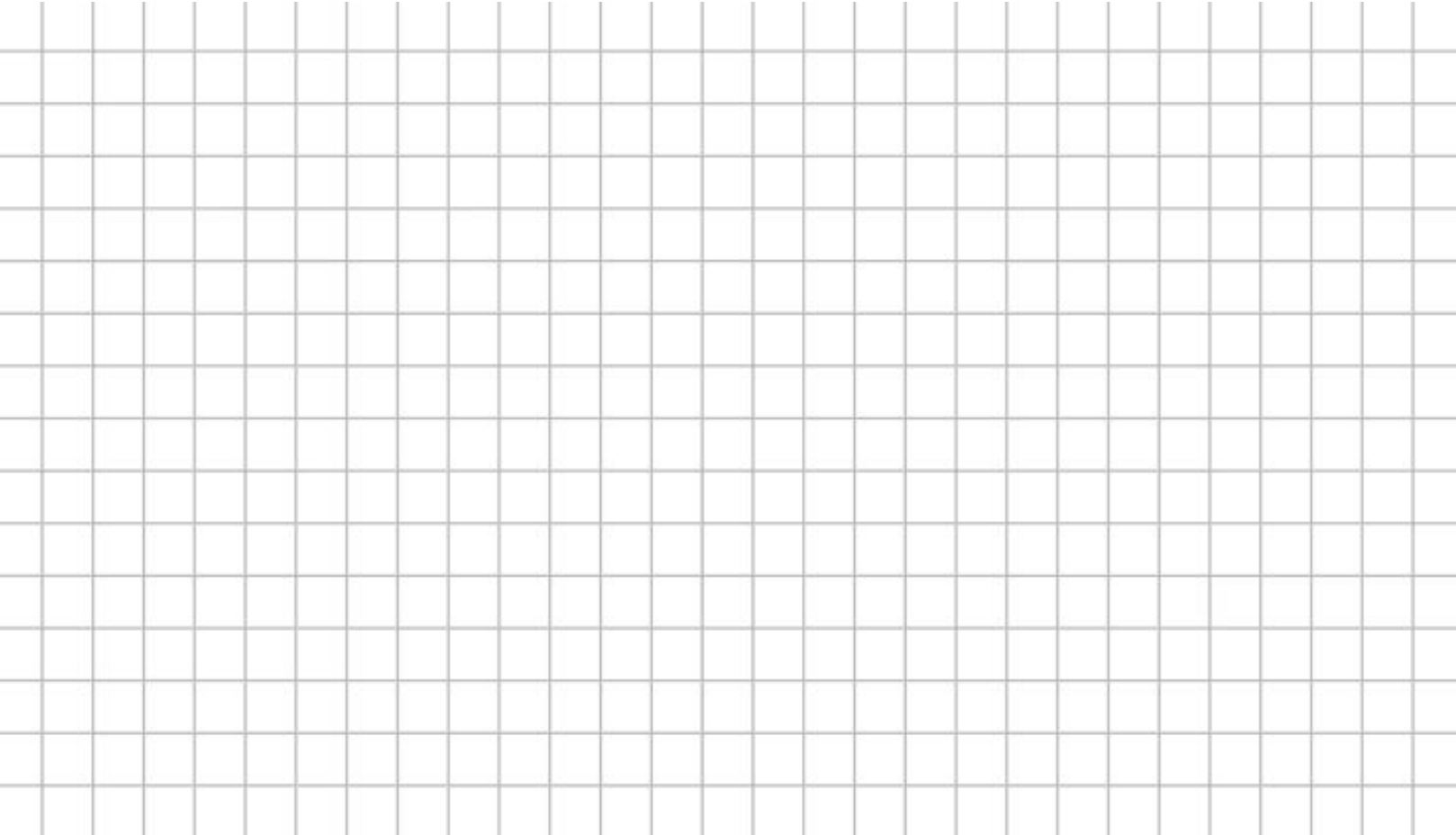
A tree drawn on the board is useless to a computer.

Until you tell the computer *how to store it*, your tree doesn't exist.

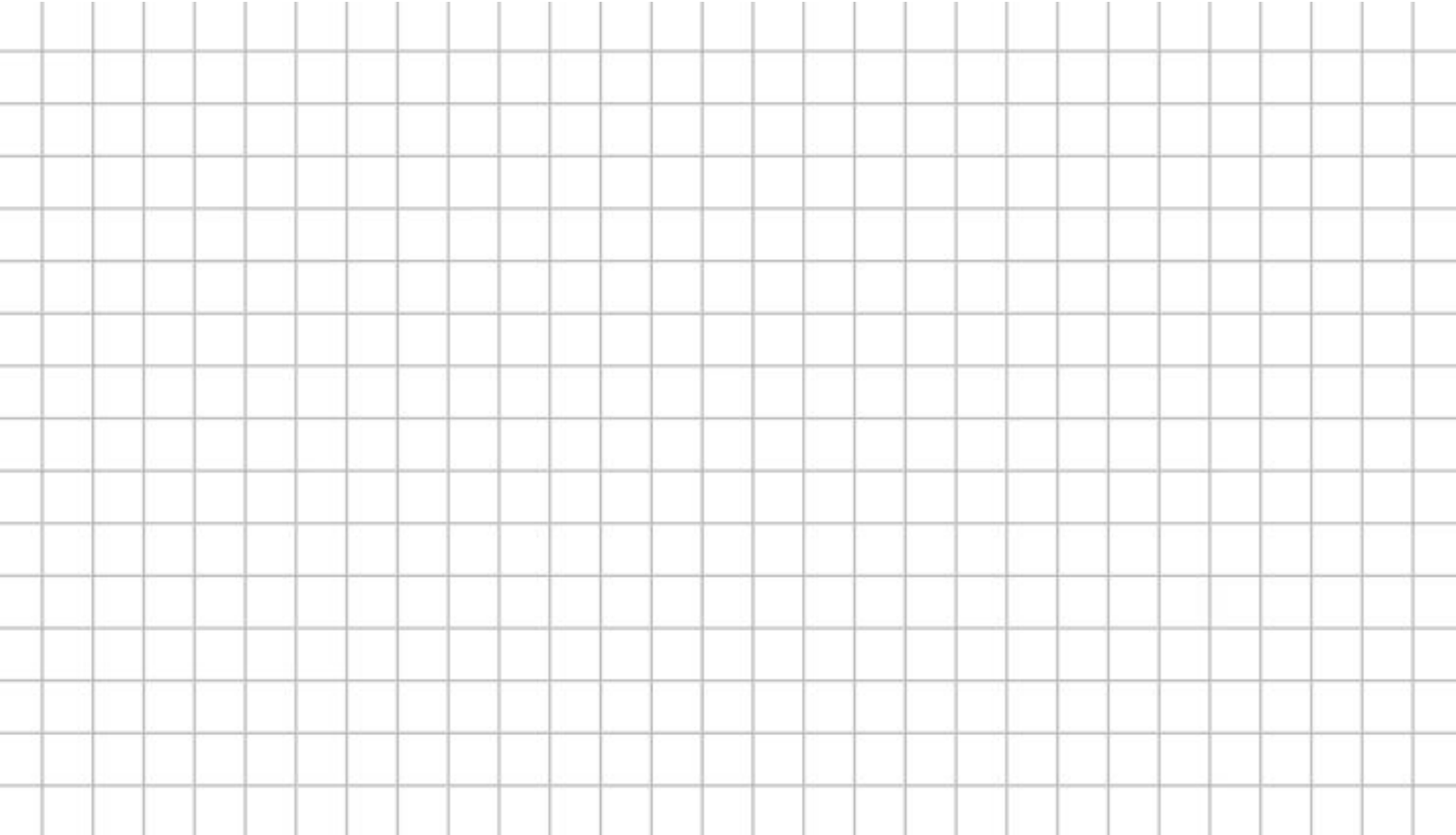
How do you dump a tree in a linear array of blocks?



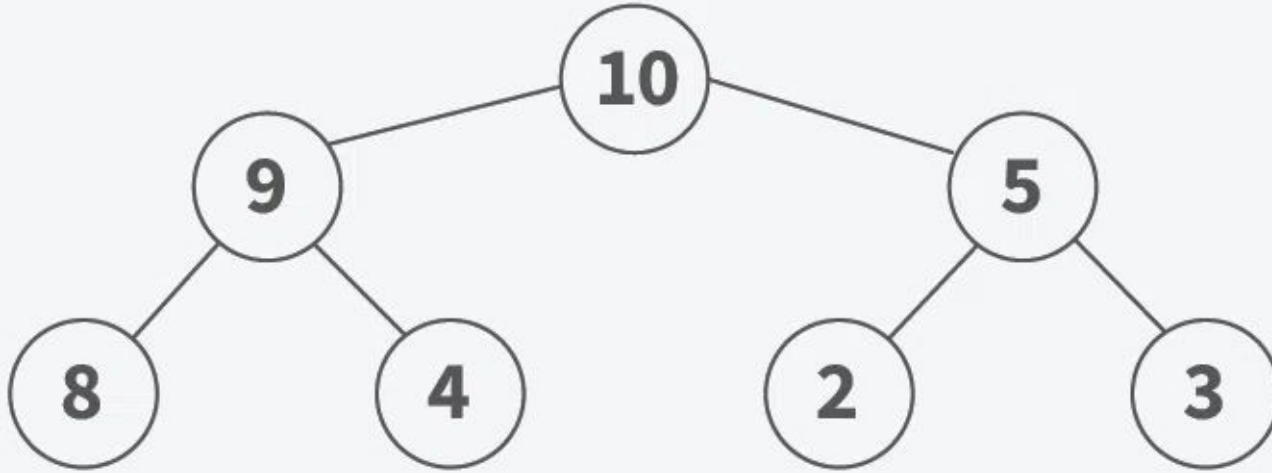
Linked List in a Computer Memory



Tree in a Computer Memory

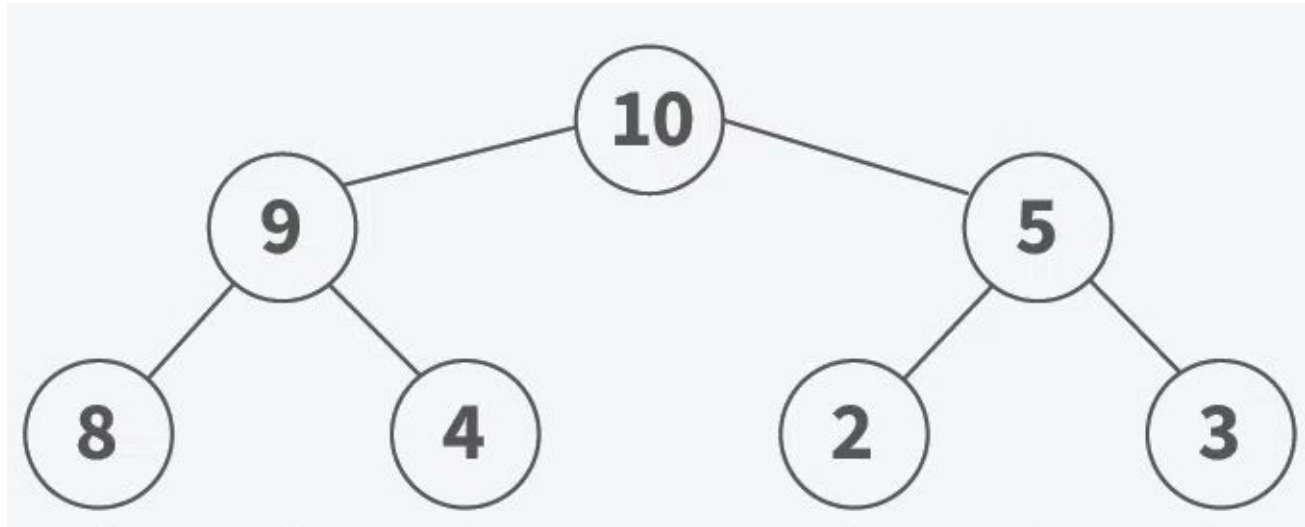


Linked Representation for Trees



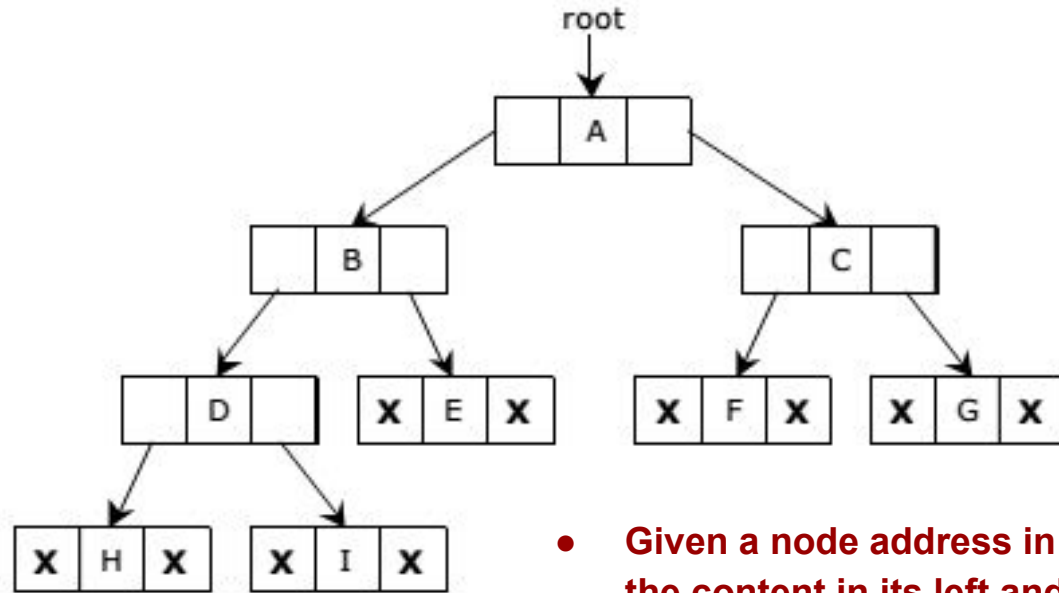
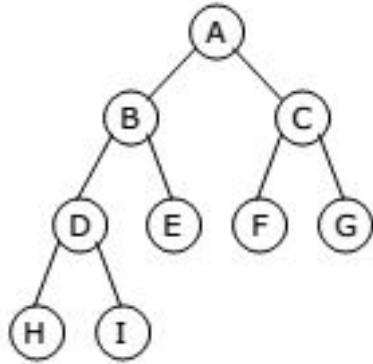
```
// Linked List  
struct node {  
    int data;  
    struct node* next;  
};
```

Linked Representation for Trees



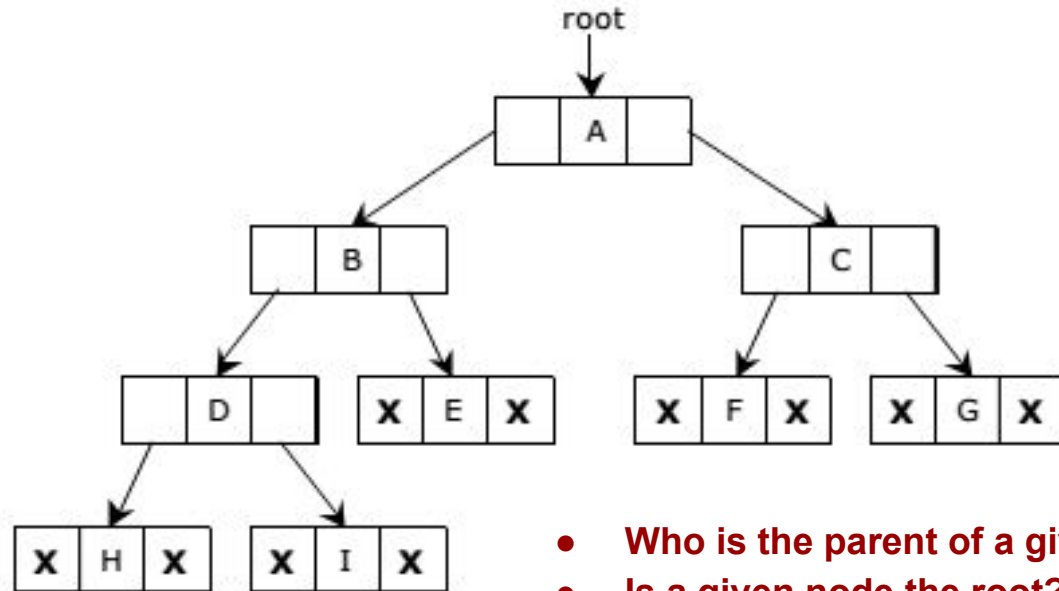
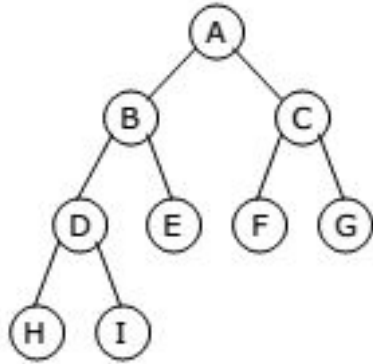
```
node* root= (node*) malloc(...); // blah blah
```

Linked Representation of a Binary Tree



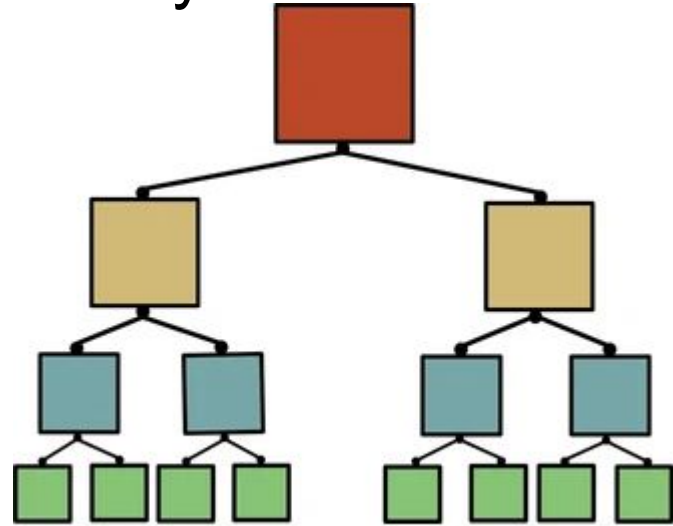
- Given a node address in a pointer ptr, find the content in its left and right children?
- Is a given node a leaf node?

Linked Representation of a Binary Tree



- Who is the parent of a given node?
- Is a given node the root?
- What is the level (depth) of a given node?
- What is the height of the tree?

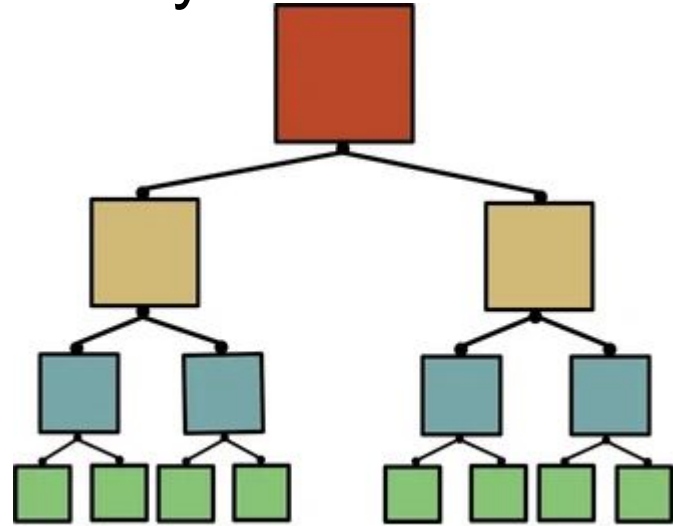
Array based Representation of a Binary Tree



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
---	---	---	---	---	---	---	---	---	---	----	----	----	----	----

Array based Representation of a Binary Tree

- Given a node index, find the index/content in its left and right children?
- Is a given node a leaf node?
- Who is the parent of a given node?
- Is a given node the root?
- What is the level (depth) of a given node?
- What is the height of the tree?



0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
---	---	---	---	---	---	---	---	---	---	----	----	----	----	----

In an array representation of a binary tree (1-based indexing), if a node is at index i , its left and right children are:

left = _____

right = _____

Constructing Trees [Input and Output?]

